

7.3 Classwork: Graphs of Logarithmic Functions (no calculator)

1. Warmup

(a) Evaluate each of the following.

(i) $\log_4 32$

(ii) $\log_8 4$

(iii) $\log_{1/2} 8$

(iv) $\log_3 \sqrt[3]{9}$

(b) Let $a > 1$ and $b > 0$. Write each expression in terms of $\log_a b$.

(i) $\log_a(a^2b)$

(ii) $\log_a\left(\frac{b}{a}\right)$

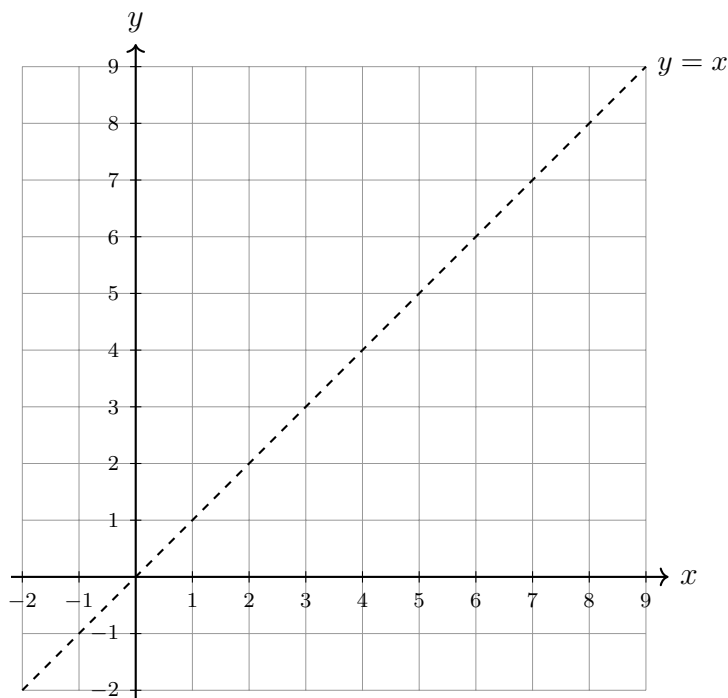
(iii) $\log_a \sqrt{a b^3}$

(iv) $\log_{a^2} b$

2. The logarithm as an inverse

You know that $y = \log_2 x$ is the inverse of $y = 2^x$. The graph of an inverse function is the reflection of the original across the line $y = x$.

The line $y = x$ is shown below (dashed) on the axes provided.



(a) On the axes above, sketch and label the graph of $y = 2^x$. Mark and label its y -intercept. Write down the equation of the horizontal asymptote.

(b) On the same axes, sketch the graph of $y = \log_2 x$. Label it, and show its x -intercept and vertical asymptote, labeling it with its equation.

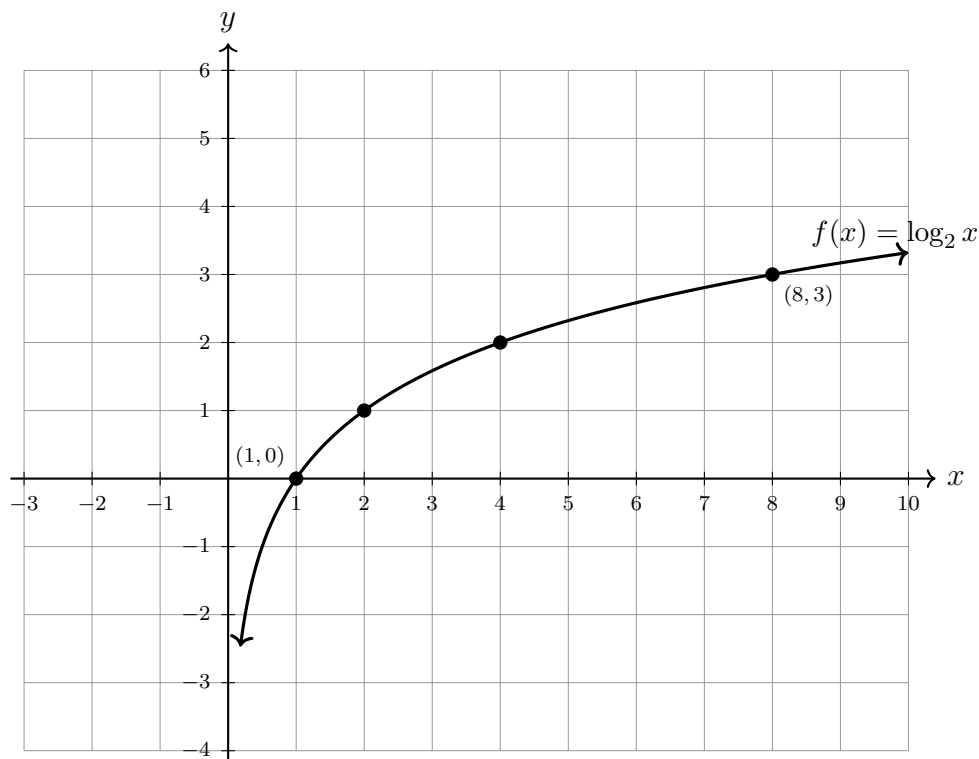
(c) Complete the table.

	Domain	Range
$y = 2^x$		
$y = \log_2 x$		

(d) The point $(3, 8)$ lies on $y = 2^x$. State the corresponding point on $y = \log_2 x$.

3. Transformations

The graph of $f(x) = \log_2 x$ is shown below, passing through $(1, 0)$, $(2, 1)$, $(4, 2)$, and $(8, 3)$, with vertical asymptote $x = 0$.



On the axes above, sketch each of the following. Label each curve and state the equation of its vertical asymptote.

(a) $g(x) = \log_2(x - 2)$ asymptote: _____

(b) $h(x) = \log_2 x + 3$ asymptote: _____

(c) $k(x) = -\log_2 x$ asymptote: _____

(d) For part (a), state the new x -intercept.

(e) $f(a) = \frac{1}{4}$. Write down the value of a . Mark this point on the graph of f as an ordered pair.

4. A logarithmic curve $y = \log_b x$ passes through the point $(27, 3)$.

(a) Find the value of the base b .

(b) State the coordinates of the point where this curve crosses the x -axis.

5. The graph of $y = \log_3(x - h) + k$ has vertical asymptote $x = 2$ and passes through the point $(5, 4)$.

(a) State the value of h .

(b) Find the value of k .

(c) Sketch the graph. Mark and label the vertical asymptote and the point $(5, 4)$.

